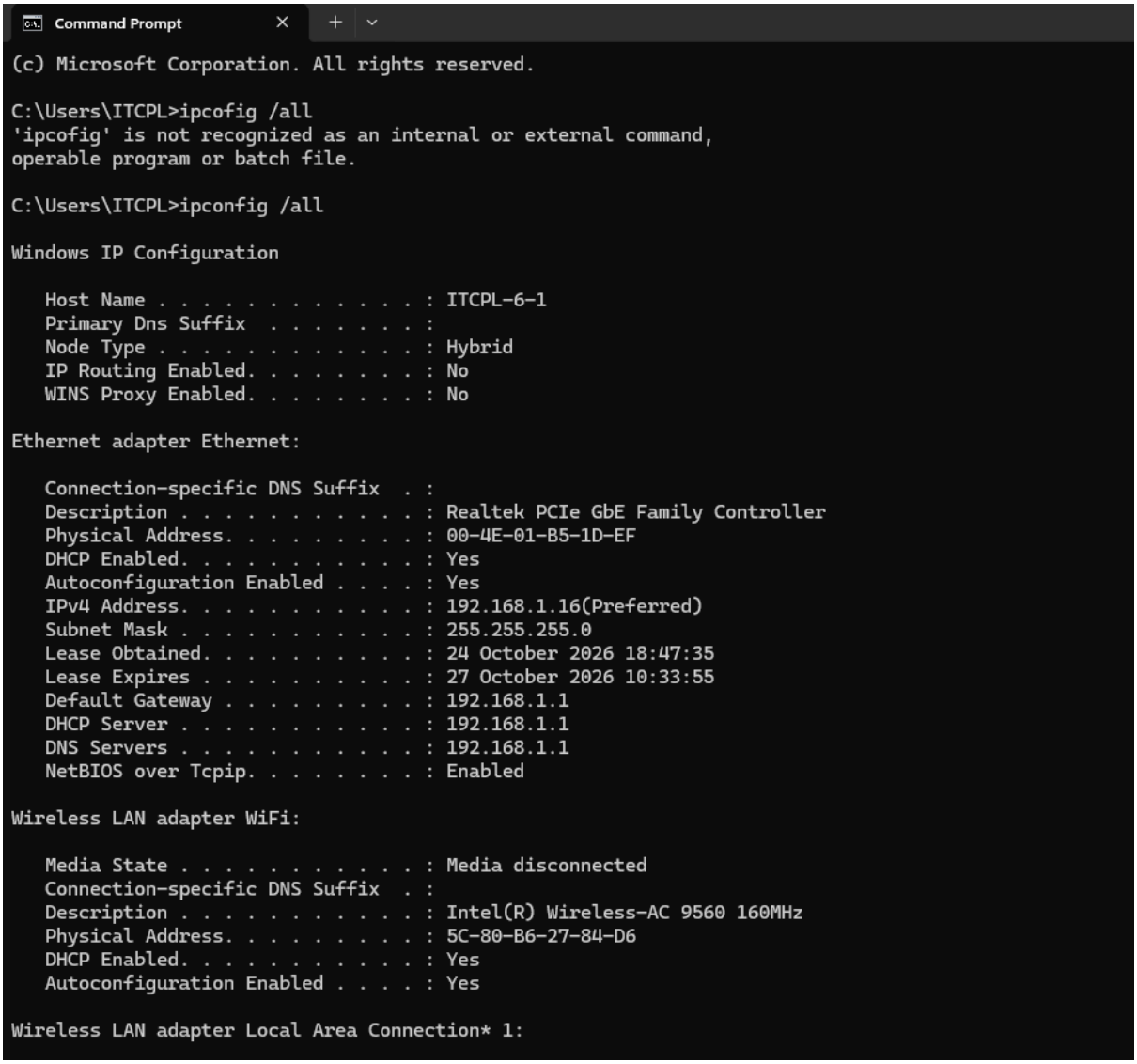
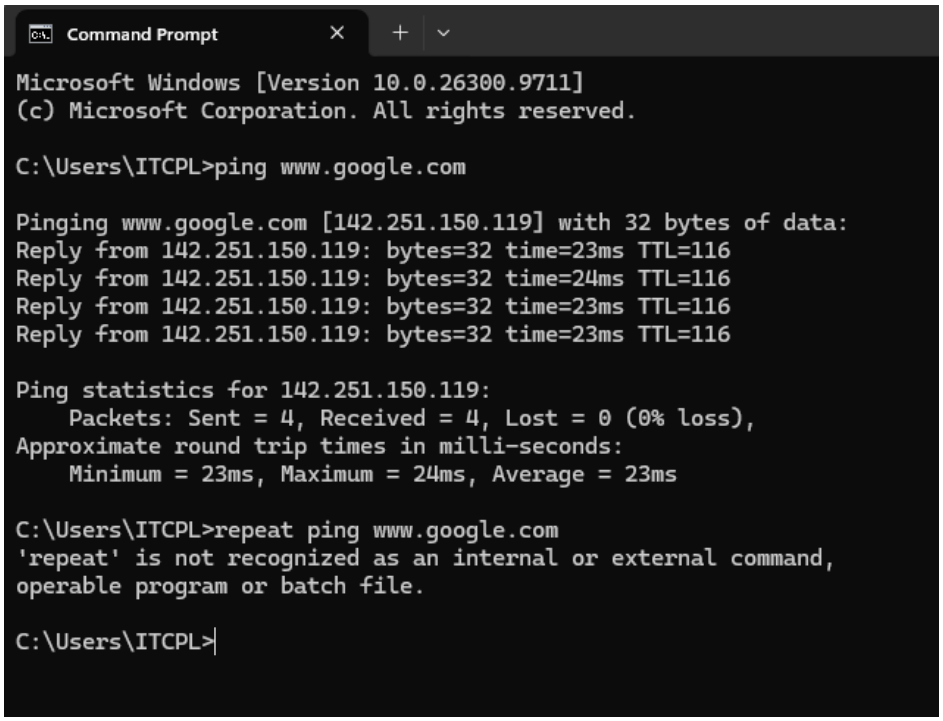
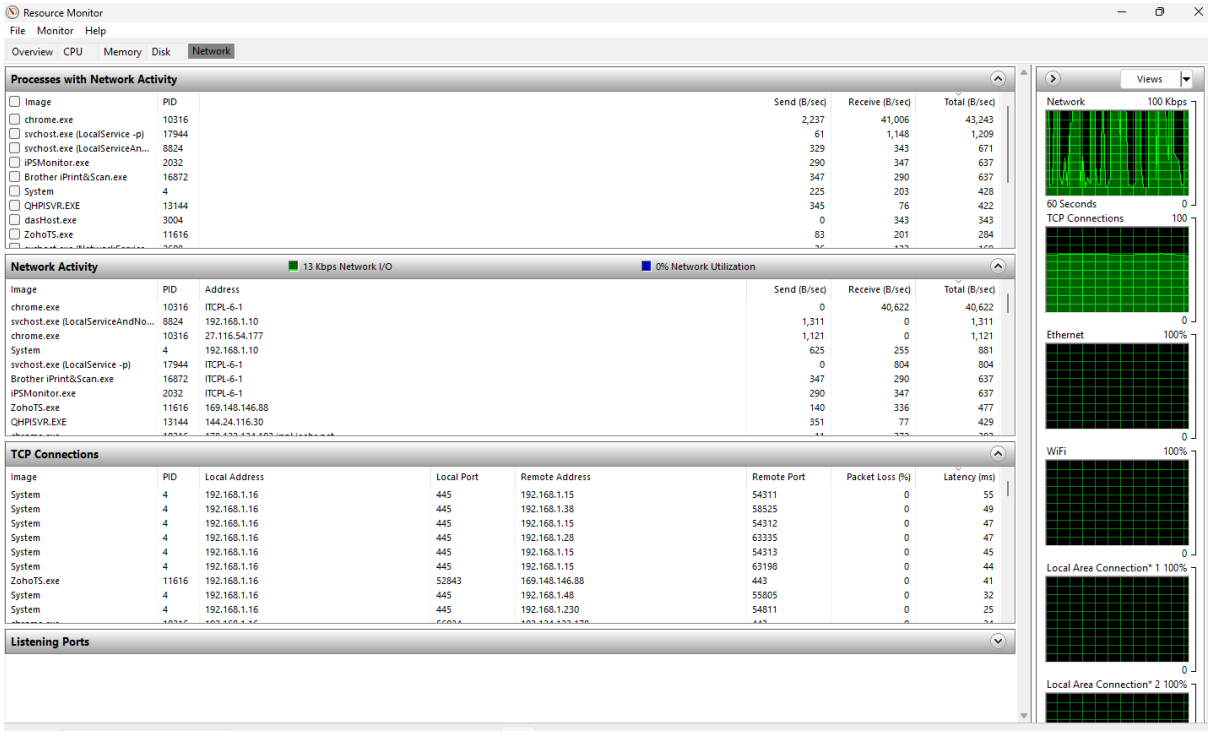
1) Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.



2) Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

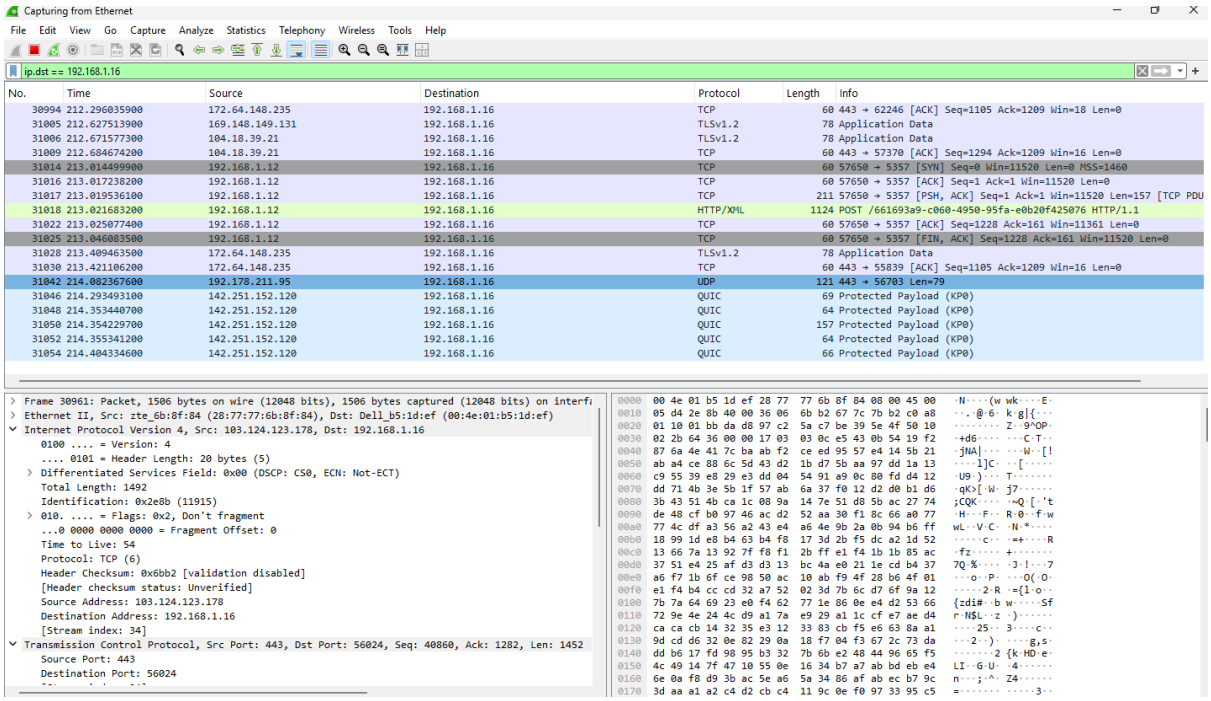


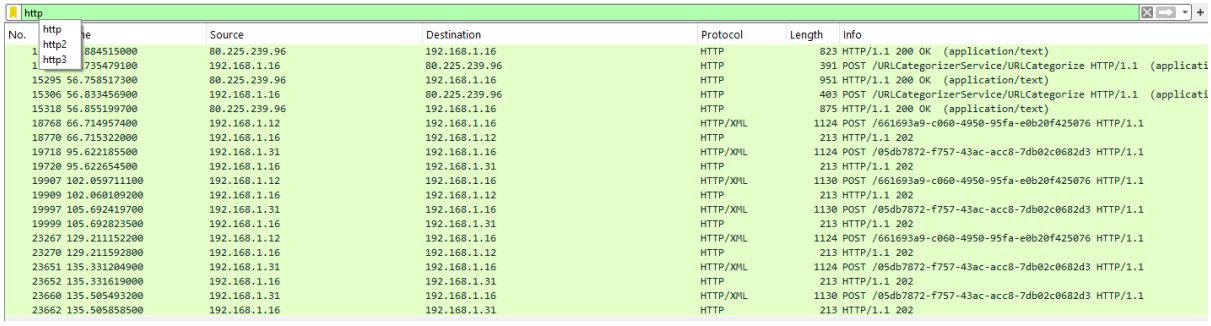
3) Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.



**4) Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.

Hint: Use the filter bar in Wireshark and try http or https as the filter keyword.**





5) AI-Generated Troubleshooting Checklist: Slow Internet While Gaming

Problem: My friend experiences slow internet, high ping, and lag while playing online games.

Troubleshooting checklist

1. Use an Ethernet cable: Connect the gaming PC or console directly to the router for a more stable connection and lower latency.

2. Restart the router: Turn off the router for about 30 seconds, then turn it back on to resolve temporary network issues.

3. Stop background downloads: Pause Windows updates, game downloads, cloud backups, and other activities that consume bandwidth.

4. Improve Wi-Fi signal: Move closer to the router or use the 5 GHz Wi-Fi band when nearby for better performance.

5. Check ping and packet loss: Use the ping command to check latency and packet loss to identify possible connectivity issues.

6. Limit other devices: Reduce heavy streaming, downloading, and video calls on the same network while gaming.

7. Update network drivers: Install the latest compatible Wi-Fi or Ethernet drivers to improve connectivity and stability.

8. Check router QoS settings: If supported, configure Quality of Service (QoS) to prioritize gaming traffic.

9. Choose a nearby game server: Select a server geographically closer to reduce latency.

10. Contact the ISP: If the problem continues, ask the internet service provider to check for outages, line faults, or routing problems